



ZHONGYI (JAMES) GUO

zhongyig123@gmail.com

LinkedIn: [zhongyi-guo](#)

[Personal Website](#)

(+1) 607-262-3415

Bay Area, CA

Summary of Experience

Data scientist with 5+ years applying statistical modeling and deep learning to large-scale data. Proficient in Python, R, and SQL, with experience building end-to-end analytical pipelines and communicating quantitative results to technical and business stakeholders..

Work Experience

DATA SCIENTIST I, W74 ADVISORS

03/2026 - PRESENT

- **Delivered analytical and statistical solutions** for client building energy optimization, owning work **end-to-end** from problem framing to stakeholder reporting.
- **Improved model performance by 25%** over baseline by benchmarking **XGBoost, LightGBM, Random Forest, Transformer, and DNN** architectures with **Optuna**-based hyperparameter tuning; final model achieved a CVMSE of ~ 0.2 , outperforming industry.
- Used **Claude Code** to **accelerate prototyping** and **maximize productivity**.

RESEARCH ASSOCIATE, THE GLADSTONE INSTITUTES

07/2025 - 04/2026

- **Developed an R package** implementing a permutation testing pipeline to validate differential gene expression analysis results.
- Collected and integrated **10 large-scale** public datasets, owning cleaning, QC, and EDA **end-to-end**. Applied the package to them to demonstrate its performance.
- Used **AlphaGenome**, a **transformer**-based model, to investigate disease pathogenesis.
- **Communicated quantitative results to technical and non-technical audiences** through **storytelling**, data visualizations, presentations, and written reports.

Project Highlight

MULTI-ASSET PORTFOLIO OPTIMIZATION WITH BITCOIN, 2026 HKU SUMMER CAMP 04/2026 - 05/2026

GITHUB REPO: <https://github.com/JG1ANDONLY/bitcoinAllocation#>

- Built an **end-to-end Bitcoin allocation framework** in Python **across 4 ETFs and BTC-USD** ($\sim 2,900$ trading days from Yahoo Finance) with aligned calendars.
- Ran 5 long-only **mean-variance optimizations** (max-Sharpe, 10%/15% vol caps, 5% daily CVaR, BTC-capped max-Sharpe); **adding BTC lifted max-Sharpe by +29%**.
- Implemented **Black-Litterman** with a market-cap prior with three explicit views, feeding posterior returns into a max-Sharpe MVO to **produce a stable 5-15% BTC weight**.
- **Quantified BTC's post-2017 regime shift** (CAPM beta vs. SPY: ~ 0 to ~ 1 , correlations ≤ 0.22 with other assets), supporting a **5-10% quarterly-rebalanced allocation**.

Education

Stanford University – MS in Epidemiology & Clinical Research (**GPA: 3.86/4.00**) 2025

Cornell University – BS in Statistics and Biology (**GPA: 3.57/4.30**) 2023

Case Western Reserve University – Biochemistry Major (**GPA: 4.0/4.0**) 2020

Skills

- Tools: R, Python, SAS (Base certified), Excel, JS (D3.js), LaTeX, HTML, CSS, Claude Code
- Aesthetics: Adobe (Illustrator, Photoshop, After Effect), Procreate

Leadership

ELECTION AIDE OFFICIAL, COUNTY OF SANTA CLARA

04/2026 - 06/2026

- Assisted with the 2026 California Primary Election voting in the county of Santa Clara.
- Guided voters through required forms and materials using the **Disney H.E.A.R.D. service model** to ensure an efficient, well-organized voting process.

EPIDEMIOLOGY M.S. STUDENT AMBASSADOR, STANFORD UNIVERSITY

05/2024 - 06/2025

- Served as **primary point of contact** for prospective applicants, fielding inquiries on curriculum, resources, and program logistics.

SCIENTIFIC REVIEW EDITOR (EDUCATION BRANCH), COMMUNITY HEALTHED

03/2022 - 05/2023

- Validated accuracy of cited information and converted scientific jargon to plain language to ensure accessibility in weekly newsletters.

ORIENTATION LEADER, CORNELL UNIVERSITY

08/2021 AND 01/2022

- Conducted icebreakers and guided campus tours to welcome new students.

Teaching Experience

CORNELL UNIVERSITY

01/2021 - 05/2023

- **Beta Tester** and **Teaching Assistant**, Introduction to Data Science
- **Grader**, Probability Models & Inference
- **Teaching Assistant**, Laboratory in Genetics and Genomics

Publication

[1] (Submitted to Nature Neuroscience, 2025) Shin, J.‡, Brady, E.‡, Chen, C., Lauderdale, K., Agrawal, A., Zhang, Y., Jiang, X., Nambiar, P., Herbert, J., Mallen, D., Ly, K., **Guo, Z.**, Sant, C., Thomas, R., Miller, S., Cobos, I., Palop, J.. APOE4 and Aβ synergize to drive neuronal network dysfunction and lysosomal-ER proteostasis dysregulation in the preclinical stages of Alzheimer's disease.

Manuscripts in preparation:

[1] (In Progress; Aiming for BMC Medicine) **Guo, Z.‡**, Chen, D.‡, Stopsack, K. H., Soule, P., Ajit, D., Ramamoorthy, P., Hoffmann, T. J., Chan, J. M., Mucci L. A., Graff, R. E.. Metabolomic Disparities Between Black and White Men with Metastatic Hormone-Sensitive Prostate Cancer: A Pilot Study.

[2] (In Progress; Aiming for Cell) Qu, P., Wang, T., Jessa, S., **Guo, Z.**, Guo, H., Purmann, C., Monte, E., Jiang, L., Yang, X., Zhou, B., Kundu, S., Kundaje, A., Wong, W., Hallmayer, J. F., Urban, A. E., Snyder, M. P.. Multi-modal functional genomics analysis of bipolar disorder and schizophrenia. (Title is tentative.)

[3] (In Progress; Aiming for Nature Neuroscience) Sant, C., **Guo, Z.**, Corces, M. R.. Preventing false discoveries in Alzheimer's disease single-cell sequencing data using permutation testing. (Title is tentative.)

[4] (In Progress; Aiming for Nature) Ee, R., Amouzgar, M., ..., **Guo, Z.**, ..., Bendall, S.. APOE4 Drives a Uniquely Dysfunctional Human Microglial State in Alzheimer's Disease. (Author list is tentative.)

‡ indicates co-first authorship.